From Refining Capacity to Import Dependence: Portugal's Diesel and Gasoline Market, 1990–2024

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Abstract

Portugal's refining system gained diesel conversion capability when the Sines hydrocracker entered commercial production in 2013, and lost one of its two refineries when refining ceased at Matosinhos in May 2021. Using monthly JODI physical flows, the Eurostat oil balance for Portugal and Spain, DGEG national statistics and the European Commission Weekly Oil Bulletin, this paper measures how the product balance and price exposure changed over 1990–2024. The annual balance runs from 1990, the earliest year Eurostat publishes; the monthly flows begin in 2002 and the weekly prices in 2005, each limited by its own source.

Over thirty-five years the system crosses the same line three times. Portugal was a net diesel exporter through most of the 1990s, a net importer for the fourteen years from 1999, an exporter again from 2013, and an importer in every year since 2021, so the 2013 hydrocracker restored a position the country had held before rather than creating a new one. Three breaks mark the crossings. Portuguese refinery output breaks upward around 2000 for both products, most strongly for gasoline ($F = 22.0$, Benjamini–Hochberg adjusted $p < 0.001$), while demand grew faster still, so the period reads as capacity chasing consumption rather than a change in orientation. The 2013 hydrocracker coincides with a second and sharper break: diesel exports and the net-import-to-demand ratio both break at 2013 (adjusted $p < 0.001$) and Portugal moved from net diesel importer to net diesel exporter. After May 2021 the direction reverses, and by 2023 domestic manufacturing covered 0.68 of diesel demand, the lowest value in the window, while imports reached a series high of 1,632 kt. Six of twenty-four break tests survive multiplicity adjustment.

Two annual designs disagree about 2022, and the disagreement is diagnostic rather than troubling. Chow tests detect nothing on three post-event observations; an interrupted-trend model fitted over the full thirty-five years detects a smaller level shift in diesel exports of -782 kt ($p = 0.022$). The 2022 estimates are considerably weaker on the long series than on a window beginning in 2005, and the gasoline ones are no longer significant, so the annual evidence for 2022 is thinner than a short window suggests. A monthly segmented model with month fixed effects separates the May 2021 closure from the March 2022 energy shock and finds both. Each phase is measured against the same counterfactual, the pre-closure trend extrapolated forward, rather than against the phase before it: diesel refinery output sits 103 kt per month below that counterfactual during the transition and 148 kt per month below it during the stress period, with the net-import-to-demand ratio 0.19 and 0.37 above it.

Over 2018–2024 Spanish diesel output covered 0.85–0.93 of Spanish demand with no trend, while Portugal's fell to 0.68. That weakens a purely common-shock account, though the two systems differ on margins this paper does not measure. On prices, both pairs are cointegrated in logs and both error-correction speeds more than triple: diesel from -0.133 to -0.466 per week and gasoline from -0.091 to -0.315. The diesel contemporaneous elasticity also rises, from 0.713 to 1.242; the gasoline elasticity does not move. Both fuels became more tightly linked to Spain, diesel through both channels and gasoline through the speed of adjustment alone. These are co-movement measures, not causal pass-through.

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The paper reports associations around documented events. It does not claim a causal effect of the closure: the transition and the largest European energy shock in decades fall ten months apart, and no design here separates their contributions.

Contents

1 Introduction and Research Question

Portugal operated two refineries for most of the study period. Sines, the larger, received a hydrocracker that entered commercial production in January 2013 and was intended to raise diesel yield relative to fuel oil. Matosinhos ceased refining in May 2021 following a December 2020 announcement, concentrating Portuguese refining at Sines. Ten months later, in March 2022, the invasion of Ukraine and the subsequent withdrawal from Russian oil products disturbed European product supply generally, and vacuum gas oil, a diesel feedstock at Sines, in particular.

The empirical question is whether the loss of refining capacity increased Portugal's dependence on imported finished petroleum products and reduced supply resilience, and whether Portugal–Spain price co-movement changed after the closure.

The answer this paper arrives at is a sequence rather than a single event, and the sequence is the argument. Portugal was a net diesel exporter through most of the 1990s, in eight of the nine years from 1990. It crossed into net import at the end of that decade and stayed there for fourteen consecutive years. Refinery output breaks upward around 2000 while neither dependence ratio moves with it: the system was growing to meet demand that was growing faster, which changes its size but not its character.

In 2013 the hydrocracker changes the character. It shifts the product slate towards middle distillates and Portugal crosses back into net export, holding that position in every year from 2013 to 2018, with 2019 a single-year interruption and 2020 a brief return. In May 2021 Matosinhos closes and the crossing reverses a third time. Every year since has been a net-import year, and in 2023 the net-import ratio reaches 0.262, the highest in the window, while domestic manufacturing covers 0.678 of diesel demand, the lowest. Ten months after the closure the European energy shock arrives and tests the reduced system, and the test falls almost entirely on diesel.

Read as a sequence, these are not three unrelated breaks but one system crossing and re-crossing a single line. That framing matters for what the 2013 unit did: it did not move Portugal somewhere new, it restored a trade position the country had held in the early 1990s. And it matters for what followed, because the 2021 closure did not merely undo that restoration. It carried the system past the dependence it had run in the 2000s, to the most import-dependent year in thirty-five. What the paper cannot do is apportion that final movement between the closure and the shock, because they fall ten months apart. That limit is why the monthly design exists and why the causal question is left open.

The paper does not assume that closure raised prices. It separates four mechanisms and tests each in turn:

refining capability → production and trade balance → physical import dependence → price co-movement.

Four hypotheses were pre-specified before estimation:

1. the 2013 hydrocracker changed diesel-producing capability and the trade balance;
2. the 2021 Matosinhos transition changed import dependence and domestic output coverage;
3. 2022 is a distinct supply-system stress episode rather than clean post-treatment evidence;
4. any price change must be evaluated on pre-tax prices, since tax changes would otherwise be read as pass-through.

A fifth candidate break year, 2000, was added after the annual panel was extended back to 1990 and the 1990s were seen for the first time. It was not pre-specified, and every result concerning it is exploratory. It is reported because omitting a visible feature of the series would be worse than labelling it, but it carries less evidential weight than the pre-specified events and no conclusion here depends on it.

Throughout, four evidence levels are kept distinct: a *documented event*, a *descriptive change*, a *statistical association or structural break*, and a *causal effect*. Only the first three are claimed anywhere in this paper.

2 What Resilience Means Here

In this study, resilience means the capacity of Portugal's domestic refining and its trade channels to absorb a product-specific supply shock without a sharp increase in import reliance or in exposure to prices formed elsewhere. That definition has three moving parts, and each is measurable here: how much of demand domestic manufacturing can cover, how far the trade channel has to stretch when manufacturing falls short, and how much a system leaning on trade gives up in price independence. A system is more resilient on this reading when a shock of a given size moves those quantities less.

The definition is deliberately narrow, and the narrowness is the reason it can be tested. It asks about the observable response of a physical balance and a price relation, not about whether the lights stay on. Four quantities carry it, and the rest of the paper is organised around them.

Domestic supply buffer. The share of consumption the country manufactures, measured as refinery output over domestic demand. This is a capability measure and not a self-sufficiency measure: output may be exported while demand is met by imports, which is the normal state for Portuguese gasoline. A ratio above one says the system *could* in principle cover its own demand from domestic manufacturing, not that it does.

Import exposure. How much of consumption arrives through the trade channel, and in which direction on balance. Gross import dependence, imports over demand, measures the volume that must be sourced abroad. The net-import-to-demand ratio measures the country's position: negative for a net seller, positive for a net buyer. The two answer different questions and a system can move on one without moving on the other, which is what happens around 2000.

Concentration. How many places the manufacturing happens in. Portugal moved from two refineries to one in May 2021. This is a documented event rather than an estimated quantity, and it is the reason the 2022 episode is examined separately: a shock to a single-site system is not the same object as the same shock to a distributed one. What that difference consists of is set out in Section 2.1.

Price integration. How closely, and how quickly, domestic pre-tax prices track a neighbouring market's. A system meeting more of its demand through imports has less room for domestic price formation, so integration is the price-side counterpart of physical exposure. Section 9.1 onwards measures it as contemporaneous co-movement and as the speed at which a gap against the long-run relation closes.

Several things that belong to a full account of security of supply are absent, and their absence bounds every claim made here. Compulsory stocks and commercial inventory are not observed, and stockholding is the first line of defence against an interruption. Port, storage and pipeline capacity, which determine whether an import volume can physically be landed and moved, are not observed. Nor are supplier diversification, crude slate, contractual terms, or spare capacity at the remaining site. A country that imports a large share of its diesel through deep, liquid, well-connected markets may be more secure than one that manufactures more of its own with a single vulnerable feedstock route. Nothing here can distinguish those cases.

What is measured, then, is the structural dimension: how much of demand the domestic system covers, how much arrives by import, how concentrated the manufacturing is, and how tightly domestic prices move with a neighbour's. Where the paper says dependence rose, it means those quantities moved. It does not mean that supply became less secure, which is a question the available data cannot answer.

2.1 What the closure changed, beyond capacity

The quantity that changed in May 2021 is capacity, and that is what the rest of this paper measures. But the closure also changed the shape of the system in ways a capacity figure does not express, and those bear directly on how the 2022 shock was met. What follows is structural reasoning about a documented event, not an estimate: none of it is measured here, and it is set out because the reader needs it to interpret what is.

Redundancy. With two refineries, an interruption confined to one leaves the other running, and the national balance absorbs part of the loss internally. With one, an interruption at that site has no domestic offset at any price, and the entire adjustment must run through trade. Portugal moved from the first situation to the second, and the observed 2022 adjustment did run entirely through trade.

Maintenance exposure. Refineries require periodic planned shutdowns. Two sites allow these to be staggered so that some domestic conversion capacity is always available; one site means every turnaround removes all of it. This is generic to refining rather than specific to Portugal, and no turnaround schedule is observed here. It matters for a different reason: an unplanned outage at Sines would leave a signature in annual balances close to the one a feedstock disruption leaves, which is why maintenance survives as an alternative explanation for 2022 that this paper cannot dismiss.

Product-balancing flexibility. Two plants of different configuration can be run to different slates, so a country can shift its product mix within domestic operations. One plant cannot: its slate is set by its own configuration and the crude available to it, and any further adjustment has to be bought or sold abroad. The configurations of the two Portuguese sites are not observed here, so how much flexibility was lost is unknown; that some was lost follows from there being one plant instead of two.

Concentration of the feedstock exposure. This is the one that matters most for what follows. The 2022 disruption was specific to vacuum gas oil, a feedstock in diesel manufacturing at Sines. Before May 2021, a VGO-specific problem at Sines would have left Matosinhos unaffected. After it, Sines was the whole of Portuguese refining, so a shock to one feedstock at one plant was a shock to national diesel manufacturing. The closure did not merely reduce capacity; it concentrated the capacity and its feedstock vulnerability in the same place, ten months before that vulnerability was tested.

This is why the paper treats the closure and the 2022 shock as parts of one chain rather than as rival explanations, and it is the substantive reason the causal question is left open rather than a statistical one.

3 Data

3.1 Sources

Four independent bodies publish the series used here. Their identifiers, roles and retrieval points were frozen at the time of extraction; full URLs appear in the References.

- **JODI Oil World Database**: monthly secondary-product imports, exports, refinery output and demand. This is the main physical-flow source because it permits transparent time aggregation and carries per-observation assessment codes.

- **Eurostat `nrg_cb_oil`**: annual oil supply, transformation and consumption for Portugal and Spain. This is the only source covering both countries on a common definition.
- **DGEG**, the Portuguese national statistical authority: annual product trade workbooks and a long domestic-sales series.
- **European Commission Weekly Oil Bulletin**: weekly consumer prices with and without taxes from 2005.

Galp corporate disclosures supply dated refinery events. They are used as event metadata only, never as outcome data, and the announcement date of December 2020 is deliberately not treated as the physical closure date.

3.2 Coverage

Table 1 states what each source covers over the study window.

Table 1: Source coverage, 1990–2024.

Source	**Role**	**Coverage**
Eurostat	Annual balance, PT and ES; the source for the annual panel	Complete for both countries from 1990, 280 of 280 product-flow cells each
JODI	Monthly physical flows; annual cross-check	Complete from 2002; every year carries 12 observed months and assessment code 1
DGEG trade	National trade statistics	**2019–2024 only.** Earlier annual workbooks are not published in this series
DGEG sales	Domestic market sales	Complete, 1970–2024
EC Bulletin	Weekly consumer prices	Complete. 995 Portuguese and 994 Spanish weekly observations per product, 2005-01-03 to 2024-12-30

The annual panel is built from Eurostat, which is the only source reaching back to 1990. JODI begins in 2002 and so cannot carry the annual arm; it supplies the monthly flows and serves as an annual cross-check over the years both cover. The monthly source runs to 2026-06 and is trimmed to the study window, leaving 276 months per product.

3.3 The one material coverage gap

DGEG publishes annual product trade workbooks only from 2019. For 2005–2018 the trade cross-check therefore runs against Eurostat rather than against the DGEG primary publication. This is weaker corroboration than it appears, because Eurostat's Portuguese oil data is compiled from the DGEG national submission and tracks it to a median difference of 0.00% across the 2019–2024 cells where both exist. Eurostat is best understood as a second channel for the same national statistics, not as an independent third opinion.

Domestic demand is unaffected. The DGEG long sales workbook covers the whole window, so demand carries three sources throughout.

3.4 Product definitions

Diesel is JODI `GASDIES` and Eurostat `O4671`, gas oil and diesel oil. Gasoline is JODI `GASOLINE` and Eurostat `O4652`, motor gasoline. Aviation gasoline is excluded from both. In the DGEG sales workbook, gasoline is the *Gasolinas* total, and diesel requires coloured and marked gasoil to be added to road gasoil; road gasoil alone sits about ten per cent below the JODI and Eurostat demand concept. Comparability is audited against a published product crosswalk, which flags the DGEG trade labels as requiring review before numerical reconciliation.

Differing biofuel treatment does not drive residual differences between sources. The Eurostat blended-biofuel wedge, `O4671` less `O4671XR5220B`, is 0.0 kt on exports in every year of the window and at most 4.5% on imports, so blending cannot explain any export divergence at all.

3.5 Cross-checking the sources against each other

Two of the three trade sources are genuinely independent, and where they disagree the paper reports the result on both. Nine of eighty JODI–Eurostat cells and eight of twenty-four JODI–DGEG cells diverge beyond tolerance. The tolerances, the divergent cells and the reason each is accepted are set out in Appendix B, and no divergent cell enters a headline figure without the alternative value being reported alongside it.

4 Methods

For fuel j in year t, with imports M, exports X, domestic demand D and refinery output Q^{refinery}:

$$\text{NetImports}_{j,t} = M_{j,t} - X_{j,t}, \qquad \text{GrossImportDependence}_{j,t} = \frac{M_{j,t}}{D_{j,t}},$$

$$\text{NetImportToDemandRatio}_{j,t} = \frac{M_{j,t} - X_{j,t}}{D_{j,t}}, \qquad \text{RefineryOutputToDemandRatio}_{j,t} = \frac{Q^{\text{refinery}}_{j,t}}{D_{j,t}}.$$

The refinery-output-to-demand ratio is *not* self-sufficiency. Portuguese refinery output may be exported while domestic demand is met partly by imports, and for gasoline that is the normal state of the system.

Annual breaks. Chow tests on a linear trend at candidate break years, of which 2013 and 2022 were pre-specified and 2000 is exploratory, with 2021 excluded as a transition year so that it is assigned to neither side, and Benjamini–Hochberg adjustment across all twenty-four tests.

Annual interrupted trends. Ordinary least squares with a common trend, a post-event level term and a post-event slope term, HAC(1) covariance, 2021 again excluded. This design retains all thirty-five annual observations, less the excluded transition year, rather than splitting the series.

Monthly event timing. A segmented regression on 276 monthly observations, 2002–2024, with month fixed effects, a calendar-month trend and HAC(3) errors. Each phase-trend interaction is centred on its phase boundary, so a phase coefficient is the level shift at that boundary against the extrapolated pre-closure trend, and the companion trend term is the change in slope within the phase. The two must be quoted together.

Prices. Model family is selected from stationarity and cointegration diagnostics rather than assumed. HAC(8) covariance throughout. Pre-tax prices are the primary measure.

5 Descriptive Evidence

5.1 Summary statistics

Over 1990–2024, Portuguese diesel demand averaged 4,522 kt against average refinery output of 4,348 kt, average imports of 804 kt and average exports of 643 kt. The two fuels sit in opposite positions relative to demand, which Section 5.4 takes up once both panels have been shown.

Peaks are informative about timing. Diesel refinery output and exports both peak in 2015, two years after the hydrocracker. Diesel imports peak in 2023, two years after the closure. Diesel demand peaks in 2004, before either event, which is the first indication that the demand path and the capacity path are not the same story.

5.2 Diesel

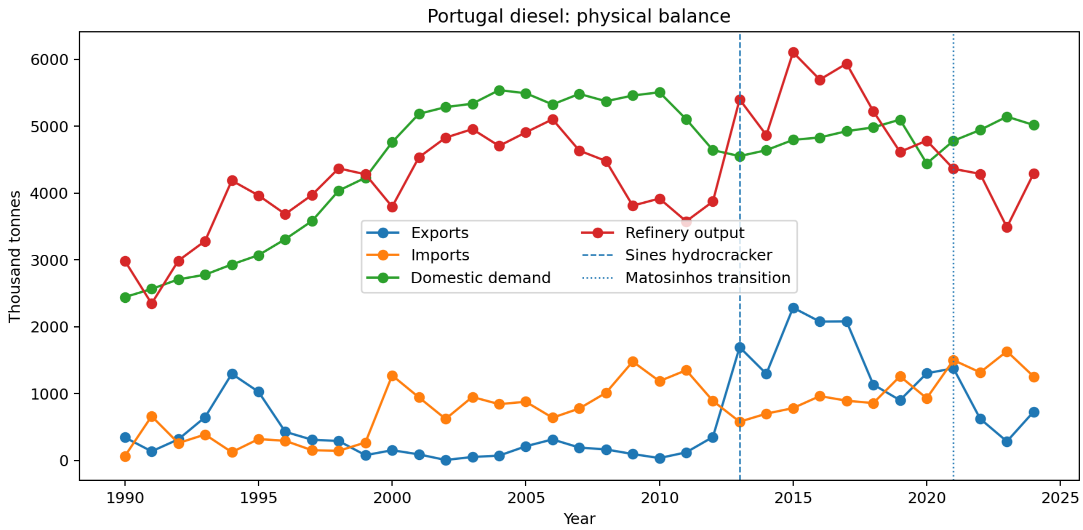

Figure 1: Portuguese diesel imports, exports, refinery output and demand, 1990–2024.

Table 2 gives the annual diesel panel, and the crossings described in Section 1 are visible in the last two columns. Taking the later part of the window, from 2005 to 2012 Portugal is a net diesel importer with output covering 70 to 96 per cent of demand. From 2013 the ratio exceeds one and the net-import-to-demand ratio turns negative: Portugal manufactures more diesel than it consumes and exports the surplus. From 2021 the system returns to net import dependence, and by 2023 output covers only 68 per cent of demand.

5.3 Gasoline

Gasoline behaves differently (Table 3). Portugal is a net gasoline exporter in every year of the window except 1993, when imports briefly exceeded exports and the net-import ratio reached +0.06. Output covers between 0.94 and 2.66 times domestic demand, the low point being that same year. From 1994 onward the ratio never falls below one, and there is no regime change comparable to diesel's: it rises after 2013 and drifts down after 2021 without approaching one again. This is precisely the asymmetry the ratio definition warns about. A system can manufacture a large surplus of one light product while importing another, and reading the ratio as self-sufficiency would obscure that.

Table 2: Portugal diesel annual balance, kt and ratios.

Year	Imports	Exports	Demand	Refinery out.	Gross imp. dep.	Net imp./dem.	Ref. out./dem.
1990	64	344	2,442	2,984	0.03	−0.11	1.22
1991	661	135	2,564	2,345	0.26	0.21	0.91
1992	259	315	2,707	2,987	0.10	−0.02	1.10
1993	387	646	2,777	3,281	0.14	−0.09	1.18
1994	123	1,295	2,931	4,186	0.04	−0.40	1.43
1995	319	1,026	3,070	3,963	0.10	−0.23	1.29
1996	293	428	3,309	3,684	0.09	−0.04	1.11
1997	152	309	3,580	3,970	0.04	−0.04	1.11
1998	143	290	4,035	4,368	0.04	−0.04	1.08
1999	268	78	4,228	4,279	0.06	0.04	1.01
2000	1,271	152	4,759	3,794	0.27	0.24	0.80
2001	943	90	5,183	4,533	0.18	0.16	0.87
2002	620	6	5,284	4,827	0.12	0.12	0.91
2003	949	51	5,335	4,955	0.18	0.17	0.93
2004	841	71	5,538	4,703	0.15	0.14	0.85
2005	877	211	5,492	4,906	0.16	0.12	0.89
2006	638	314	5,324	5,102	0.12	0.06	0.96
2007	776	192	5,482	4,634	0.14	0.11	0.85
2008	1,011	164	5,372	4,478	0.19	0.16	0.83
2009	1,478	95	5,457	3,810	0.27	0.25	0.70
2010	1,185	35	5,506	3,917	0.22	0.21	0.71
2011	1,349	120	5,103	3,576	0.26	0.24	0.70
2012	890	350	4,641	3,872	0.19	0.12	0.83
2013	578	1,693	4,551	5,399	0.13	−0.25	1.19
2014	699	1,295	4,640	4,865	0.15	−0.13	1.05
2015	782	2,284	4,794	6,104	0.16	−0.31	1.27
2016	962	2,076	4,830	5,695	0.20	−0.23	1.18
2017	893	2,079	4,924	5,935	0.18	−0.24	1.21
2018	857	1,134	4,981	5,224	0.17	−0.06	1.05
2019	1,258	900	5,097	4,614	0.25	0.07	0.91
2020	929	1,303	4,441	4,783	0.21	−0.08	1.08
2021	1,500	1,377	4,780	4,361	0.31	0.03	0.91
2022	1,316	622	4,942	4,285	0.27	0.14	0.87
2023	1,632	284	5,142	3,487	0.32	0.26	0.68
2024	1,254	728	5,017	4,291	0.25	0.10	0.86

Table 3: Portugal gasoline annual balance, kt and ratios.

Year	Imports	Exports	Demand	Refinery out.	Gross imp. dep.	Net imp./dem.	Ref. out./dem.
1990	0	294	1,370	1,704	0.00	−0.21	1.24
1991	122	252	1,507	1,623	0.08	−0.09	1.08
1992	101	244	1,692	1,860	0.06	−0.08	1.10
1993	204	103	1,795	1,687	0.11	0.06	0.94
1994	25	533	1,851	2,354	0.01	−0.27	1.27
1995	34	904	1,890	2,757	0.02	−0.46	1.46
1996	53	658	1,965	2,591	0.03	−0.31	1.32
1997	2	888	1,949	2,835	0.00	−0.45	1.45
1998	0	789	2,027	2,792	0.00	−0.39	1.38
1999	11	588	2,079	2,654	0.01	−0.28	1.28
2000	154	397	2,123	2,296	0.07	−0.11	1.08
2001	10	635	1,995	2,615	0.01	−0.31	1.31
2002	60	540	2,067	2,518	0.03	−0.23	1.22
2003	55	782	2,005	2,732	0.03	−0.36	1.36
2004	130	805	1,927	2,551	0.07	−0.35	1.32
2005	128	792	1,808	2,466	0.07	−0.37	1.36
2006	104	1,270	1,674	2,750	0.06	−0.70	1.64
2007	89	1,091	1,589	2,591	0.06	−0.63	1.63
2008	115	767	1,488	2,091	0.08	−0.44	1.41
2009	240	835	1,462	2,056	0.16	−0.41	1.41
2010	178	1,035	1,388	2,218	0.13	−0.62	1.60
2011	191	799	1,248	1,875	0.15	−0.49	1.50
2012	160	898	1,127	1,864	0.14	−0.65	1.65
2013	102	1,293	1,100	2,338	0.09	−1.08	2.13
2014	154	976	1,092	1,924	0.14	−0.75	1.76
2015	137	1,786	1,080	2,694	0.13	−1.53	2.49
2016	167	1,742	1,069	2,619	0.16	−1.47	2.45
2017	124	1,885	1,032	2,749	0.12	−1.71	2.66
2018	149	1,644	1,027	2,501	0.15	−1.45	2.43
2019	261	1,585	1,067	2,401	0.24	−1.24	2.25
2020	193	1,271	883	1,956	0.22	−1.22	2.22
2021	202	1,424	976	2,146	0.21	−1.25	2.20
2022	230	1,502	1,073	2,247	0.21	−1.18	2.09
2023	347	1,254	1,193	2,063	0.29	−0.76	1.73
2024	300	1,728	1,260	2,660	0.24	−1.13	2.11

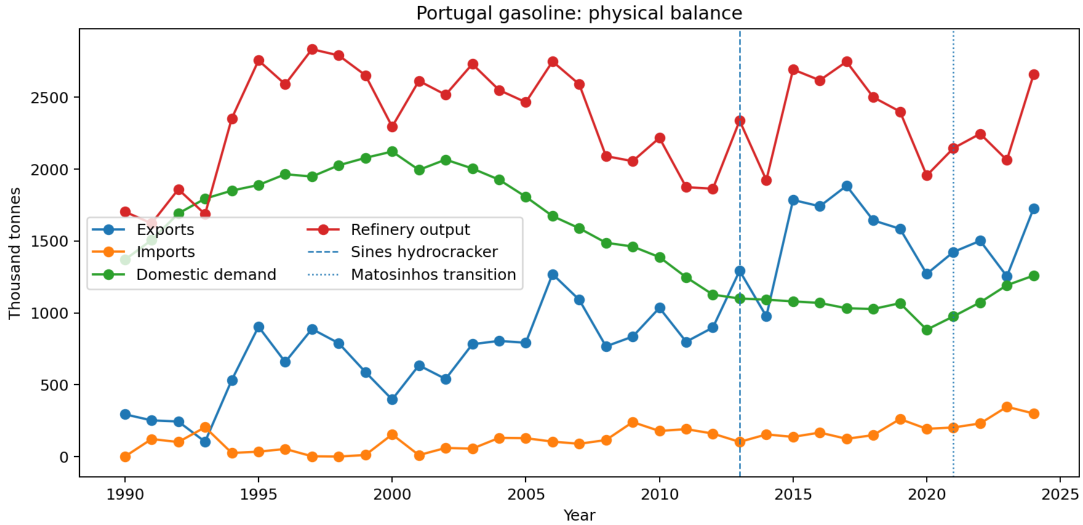

Figure 2: Portuguese gasoline imports, exports, refinery output and demand, 1990–2024.

5.4 Why the two products diverge

The two products behave differently in almost every result that follows, and the reason is visible in the window means. Portuguese diesel demand averaged 4,522 kt against gasoline demand of 1,511 kt: the country consumes roughly three times as much diesel as gasoline. Manufacturing runs the other way relative to demand. Gasoline output averaged 2,337 kt against that 1,511 kt of demand, a surplus exported at an average of 971 kt a year against imports of only 129.5 kt. Diesel output averaged 4,348 kt against 4,522 kt of demand, a deficit covered by imports averaging 804 kt against exports of 643 kt.

This is the familiar European mismatch in national form. A refinery converts crude into a slate of products in proportions set by the crude and the conversion units installed, and that slate is not free to follow demand. Portuguese demand dieselised. Gasoline demand peaks in 2000 and no later year matches it, while diesel demand peaks in 2004. The gap between the two widened over the window rather than being fixed: diesel ran at under twice gasoline demand in the early 1990s and at around four times by its end. The refining system was left producing more gasoline than the country could burn and less diesel than it needed, so gasoline became a structural export and diesel a structural import.

That difference in starting position, rather than any difference in how the two products were treated, is what makes them respond differently to the same events. A hydrocracker converts heavy gasoil into middle distillates, so the 2013 unit acted on the deficit product: diesel output peaks at 6,104 kt in 2015 and the trade position crosses into surplus. Gasoline had no deficit to close and no comparable break follows.

The same asymmetry governs the losses. Gasoline entered the 2021 closure with output well above domestic demand, so removing capacity moved the ratio down a wide margin without reaching one: the gasoline coverage ratio stays above 1.7 in every year after the closure. Diesel entered with a ratio near one and no comparable margin, so the same proportional loss pushed it to 0.678 by 2023 and turned the country back into a net buyer. A buffer is only protective while it is large relative to the shock, and Portugal held a large one in the product it did not much need and a thin one in the product it did.

One question the ratios do not answer on their own is why Portugal keeps manufacturing gasoline it does not consume, rather than making less of it. The answer is that it largely cannot. A refinery's products come out of the same barrel in proportions bounded by its configuration, so gasoline output is not an independent decision: cutting it means cutting throughput, and cutting throughput means making less diesel too. A country running its plants at the rate its diesel demand requires will produce more gasoline than it needs, and the surplus is exported because the alternative is not producing the diesel.

That inverts the natural reading of the gasoline figures. The persistent gasoline export surplus is not a sign of strength in gasoline; it is a residue of running the system for diesel. It also explains why gasoline coverage held up through the closure while diesel did not. Gasoline output tracks throughput, Sines continued to run, and gasoline demand is small enough relative to that throughput that a large proportional loss of national capacity still left output well above it. Diesel had no such cushion because diesel demand is what sets the throughput in the first place.

This account is consistent with every result reported below, but it is not tested by any of them. Yield economics, the crude slate and the commercial logic of export contracts are not observed here, and a competing account in which the two products differ for reasons of contracting rather than chemistry would fit the same series.

5.5 Dependence ratios

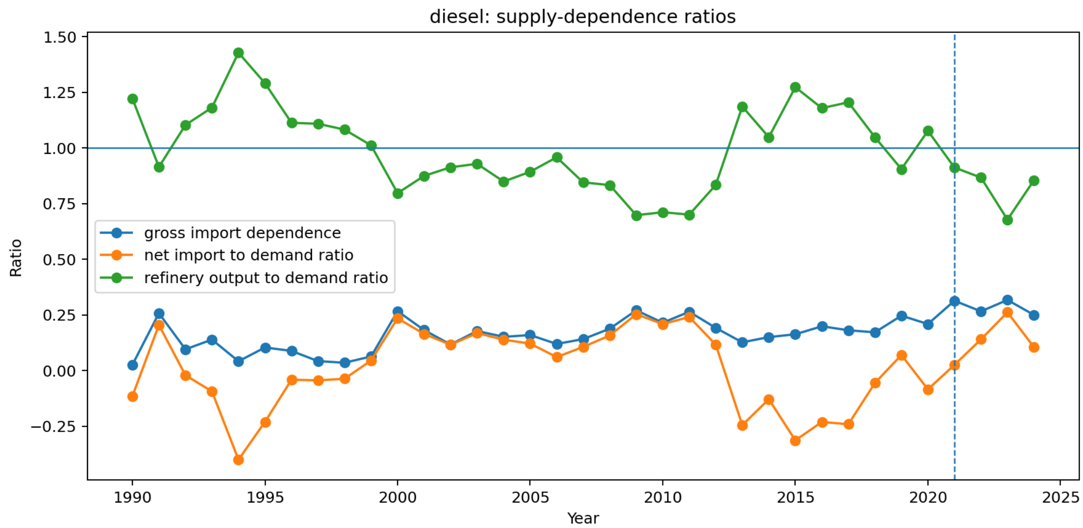

Figure 3: Diesel gross import dependence, net imports to demand, and refinery output to demand. The output ratio crossing one in 2013 and falling back after 2021 is the central physical result of this paper.

5.6 Event years

Table 4 extracts the years that bracket each documented event, together with the refining regime in force. The 2021 row is a transition year and is excluded from the annual before/after designs.

6 Structural Breaks

6.1 Chow tests

Twenty-four Chow tests are run across three candidate break years, and Table 5 lists the six that survive Benjamini–Hochberg adjustment. They fall into two groups, and only the first was pre-specified.

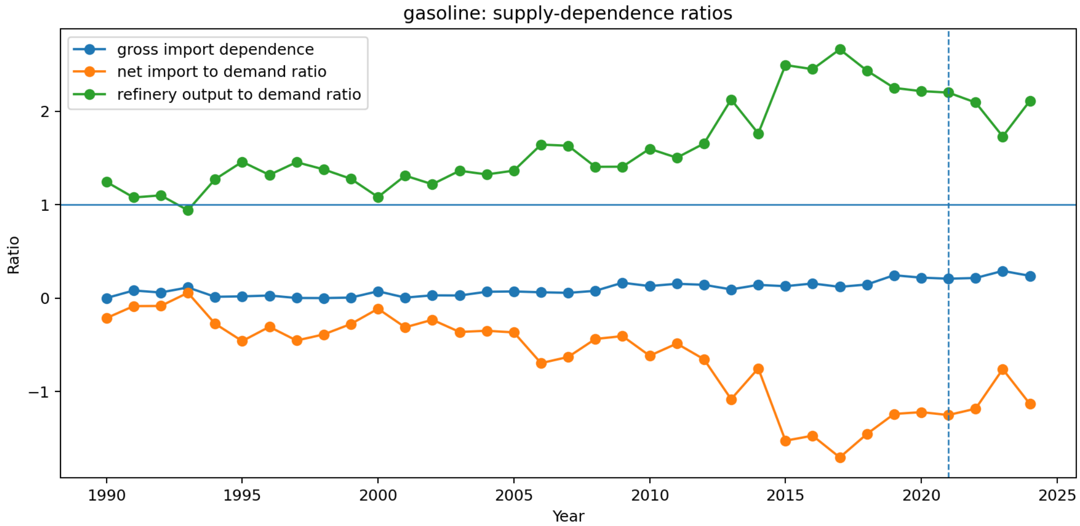

Figure 4: Gasoline dependence ratios. The output ratio dips just below one in 1993 and stays well above it in every other year, so no regime change comparable to diesel occurs.

Table 4: Balance at the event years. [†]The two independent trade sources disagree on 2022 diesel exports; the national figure is 622.2 kt, which puts net imports over demand at +0.132 rather than +0.190. See Table 15.

Year	Product	Exports	Imports	Demand	Ref. out.	Net imp./dem.	Regime
2012	diesel	350	890	4,641	3,872	+0.116	two refineries
2013	diesel	1,693	578	4,551	5,399	−0.245	two refineries
2020	diesel	1,303	929	4,441	4,783	−0.084	two refineries
2021	diesel	1,377	1,500	4,780	4,361	+0.026	transition
2022	diesel	622	1,316	4,942	4,285	+0.140	Sines only
2023	diesel	284	1,632	5,142	3,487	+0.262	Sines only
2024	diesel	728	1,254	5,017	4,291	+0.105	Sines only
2012	gasoline	898	160	1,127	1,864	−0.655	two refineries
2013	gasoline	1,293	102	1,100	2,338	−1.083	two refineries
2020	gasoline	1,271	193	883	1,956	−1.221	two refineries
2021	gasoline	1,424	202	976	2,146	−1.252	transition
2022	gasoline	1,502	230	1,073	2,247	−1.185	Sines only
2023	gasoline	1,254	347	1,193	2,063	−0.761	Sines only
2024	gasoline	1,728	300	1,260	2,660	−1.134	Sines only

Three concern 2013 and move in the direction the engineering implies: more diesel manufactured, more exported, and a net-import position that turns negative. Extending the series to 1990 strengthens them, because the pre-event segment now carries thirteen observations rather than eight.

The other three concern 2000, which is exploratory rather than pre-specified: it was added once the extended panel revealed the 1990s. Refinery output breaks upward for both products, most strongly for gasoline, and diesel imports break upward with it. What does not break at 2000 is either ratio: neither the net-import-to-demand nor the refinery-output-to-demand ratio shows a discontinuity there. Output and imports both rose because demand rose faster, so the system was expanding rather than changing character. That is a different kind of event from 2013, and treating the two alike would misread it.

Table 5: Chow tests at candidate break years, 2021 excluded as a transition year. 2013 and 2022 were pre-specified; 2000 was added after the panel was extended back to 1990 and is exploratory.

Product	Outcome	Break	F	p	BH p	$n_{\text{pre}}/n_{\text{post}}$
gasoline	refinery output	2000	22.00	< 0.001	< 0.001	10/13
diesel	exports	2013	21.58	< 0.001	< 0.001	13/8
diesel	net imports / demand	2013	21.40	< 0.001	< 0.001	13/8
diesel	refinery output	2000	11.68	< 0.001	0.0030	10/13
diesel	refinery output	2013	8.23	0.0032	0.0152	13/8
diesel	imports	2000	5.81	0.0108	0.0431	10/13

6.2 Interrupted trends, and why they disagree about 2022

The absence of a detected 2022 Chow break is specific to that test. A Chow test asks whether two independently fitted trends beat one; with three post-event annual observations it cannot answer that question. An interrupted-trend model asks whether the level shifts against a trend estimated from the whole series, a far weaker demand on the post-event data. On that specification, reported in Table 6, 2022 is detectable for diesel but not for gasoline.

The window matters here more than anywhere else in this paper. Fitted from 2005 the 2022 diesel export shift is roughly twice as large as it is when fitted from 1990 against a trend informed by the 1990s, where it is -782 kt, and the gasoline shifts lose significance altogether. The longer series is the more honest one, and it makes the annual case for 2022 weaker than a short window implies. The monthly design of Section 6 remains the stronger instrument for that period.

Table 6: Interrupted-trend models, annual series, HAC(1), $n = 34$, 2021 excluded. The 2022 export rows depend on trade cells the two sources dispute, one for each product; see Table 15.

		Level change		Slope change	
Product	Outcome, event	Estimate	p	Estimate	p
diesel	exports, 2013	+2,005.5	< 0.001	−117.1	< 0.001
diesel	exports, 2022	−781.7	0.022	+15.0	0.872
diesel	net imp./dem., 2013	−0.540	< 0.001	+0.029	< 0.001
diesel	net imp./dem., 2022	+0.201	0.031	−0.017	0.616
gasoline	exports, 2013	+411.1	0.039	−22.4	0.355
gasoline	exports, 2022	−270.9	0.121	+69.3	0.413
gasoline	net imp./dem., 2013	−0.713	< 0.001	+0.038	0.157
gasoline	net imp./dem., 2022	+0.307	0.095	+0.070	0.457

The ‡ marks the 2022 gasoline export row, which does not reach five per cent under either trade source once the panel is extended, as Section B.2 sets out. It is not relied on in the conclusions.

The two designs are not in conflict about the world; they are in conflict about what three annual observations can support. Any statement that annual evidence detects no 2022 break should therefore be read as specific to the Chow specification. The interrupted-trend models and the monthly design of Section 6 agree with each other in sign and rough magnitude.

7 Monthly Event Analysis

7.1 Phase means

The monthly panel is partitioned into four phases: before May 2021, the Matosinhos transition from May 2021 to February 2022, the energy-stress period from March 2022 to December 2022, and post-stress from January 2023. Table 7 gives diesel phase means.

Table 7: Diesel monthly means by event phase, kt per month unless noted.

Phase	Months	Imports	Exports	Ref. out.	Net imp./dem.
Pre-closure	232	75.4	65.4	411.5	0.013
Matosinhos transition	10	133.3	63.0	321.9	0.162
Energy stress 2022	10	111.1	27.7	324.6	0.193
Post-stress	24	117.5	42.3	311.7	0.181

Mean monthly diesel imports rise by roughly a half to three quarters between the pre-closure phase and every subsequent phase, refinery output falls by a fifth and then a quarter, and the net-import-to-demand ratio moves from approximately zero to 0.18. The phase means are descriptive and take no account of seasonality or trend; the model below does both.

7.2 Segmented model

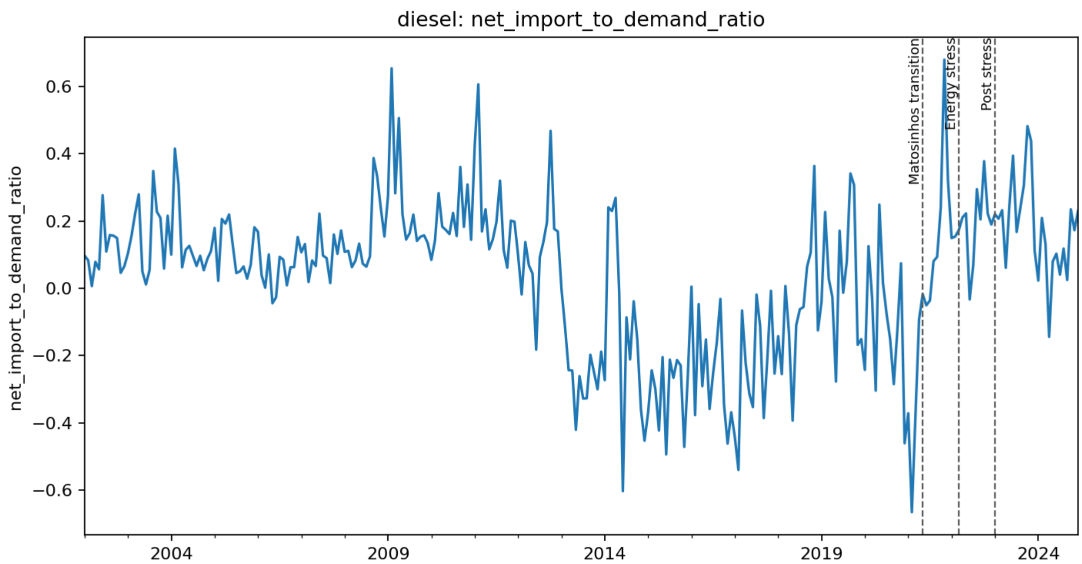

Figure 5: Monthly diesel net imports to demand, with the May 2021, March 2022 and January 2023 phase boundaries marked.

Table 8: Segmented monthly event model, diesel, $n = 276$ months (2002–2024), month fixed effects, HAC(3). A phase coefficient is the level shift at that phase boundary against the extrapolated pre-closure trend.

Outcome	Term	Estimate	Std. error	p
Refinery output (kt/month)	Matosinhos transition	−102.89	36.53	0.005
	phase trend	−4.04	6.80	0.553
	Energy stress 2022	−148.25	23.41	< 0.001
	phase trend	+5.16	3.51	0.142
Imports (kt/month)	Matosinhos transition	+40.34	23.15	0.081
	phase trend	+3.27	6.16	0.595
	Energy stress 2022	+25.43	10.49	0.015
	phase trend	+1.63	2.35	0.488
Net imports / demand	Matosinhos transition	+0.1923	0.0711	0.007
	phase trend	+0.0382	0.0152	0.012
	Energy stress 2022	+0.3658	0.0551	< 0.001
	phase trend	+0.0098	0.0072	0.171

For refinery output and the net-import ratio, both phases carry independent, statistically distinguishable effects, which is the reason a monthly design was required at all: annual data cannot separate a May 2021 event from a March 2022 one. The import level is weaker: its energy-stress term reaches five per cent ($p = 0.015$) but its Matosinhos term only ten per cent ($p = 0.081$), so the import channel is not relied on for the closure itself.

Diesel refinery output sits about 103 kt per month below the extrapolated pre-closure trend during the transition and about 148 kt per month below it during the stress period. The second figure is measured against the same counterfactual as the first, not against the transition phase, so the two do not add. The net-import-to-demand ratio sits 0.19 above that counterfactual at the closure and 0.37 above it during the stress phase, and the closure phase also carries a significant upward slope of 0.038 per month, meaning dependence continued to widen within that phase rather than stepping once and holding.

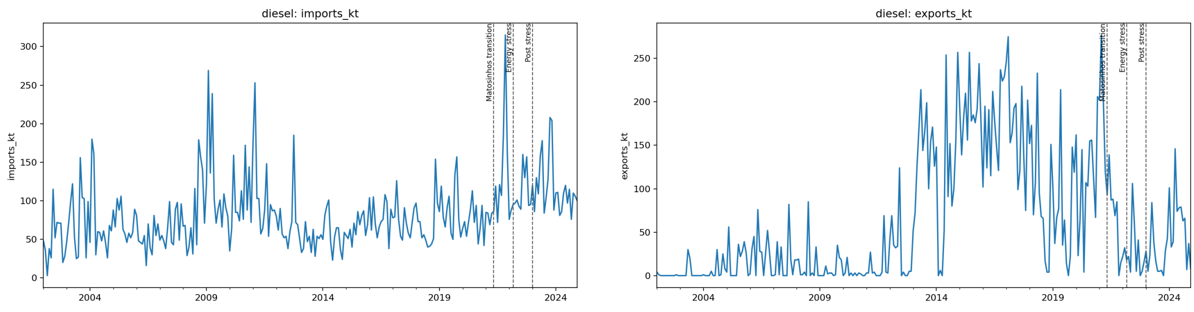

Figure 6: Monthly diesel imports (left) and exports (right).

8 The 2022 Stress Episode

The chain this section tests runs as follows. Russia supplied a large share of the world's vacuum gas oil; VGO is a feedstock in diesel manufacturing at Sines; Galp announced the suspension of Russian oil-product purchases in March 2022. If that disruption bound, diesel manufacturing should fall while gasoline manufacturing does not, because the constraint is on a diesel feedstock

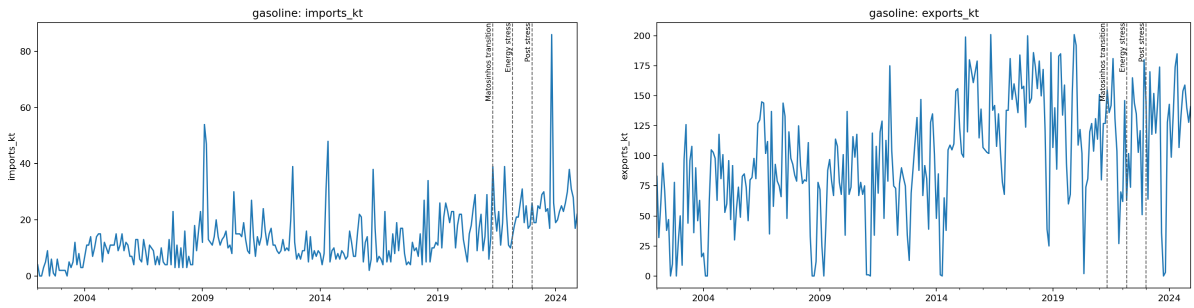

Figure 7: Monthly gasoline imports (left) and exports (right).

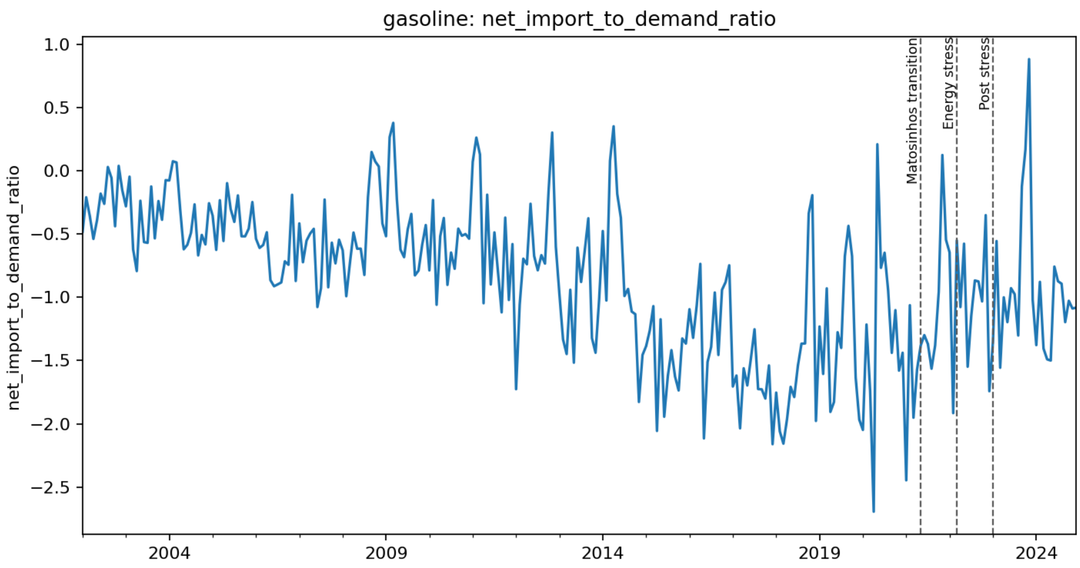

Figure 8: Monthly gasoline net imports to demand. The series sits below zero in nearly every month, with occasional single months where imports exceed exports, consistent with the annual picture of a persistent export surplus.

and not on running the plant. Domestic demand still has to be met, so exports should give way first and imports rise second. And because Portugal had been a one-refinery system for ten months, none of that adjustment can be taken up domestically. Each link is testable against the balance, and each holds.

Against a 2013–2019 baseline, 2022 diesel exports sit at the zeroth percentile of the baseline distribution and diesel imports of 1,316 kt sit 2.10 standard deviations above the mean, at the hundredth percentile. A seven-year baseline gives the standard deviation few observations to work with, so the median-and-MAD form is reported alongside it: on that measure the same figure stands at 2.94. The percentile, which assumes nothing about the shape of the baseline, is the statistic to lean on.

Table 9: 2022 diesel exports against the 2013–2019 baseline, computed on each source.

Source for the 2022 cell	Value (kt)	z	Baseline percentile
Monthly source, flagged as the outlier	331.0	−2.44	0
National statistics, corroborated	622.2	−1.89	0

Robust statistics agree. Against the baseline median and median absolute deviation, diesel imports score 2.94 and diesel net imports 3.14, both firmly outside the baseline range. Diesel refinery output in 2022 sits at $z = -2.04$ and at the zeroth percentile.

Gasoline exports in the same year are unremarkable, at $z = -0.18$ and the 29th percentile, and gasoline refinery output is only mildly low ($z = -0.76$). Gasoline imports are elevated ($z = 1.46$), but the gasoline net-import position is not ($z = 0.41$). Diesel demand in 2022 is close to baseline ($z = 0.58$), so the diesel import surge is not demand-driven.

The stress is therefore diesel-specific on the supply side. That is consistent with the documented suspension of Russian oil-product and vacuum gas oil purchases feeding diesel manufacturing at Sines, and inconsistent with a general refining shutdown, which would have moved gasoline equally. The metrics are reported across three baseline windows and the ranking is unchanged in each.

The pattern is worth reading as a whole rather than statistic by statistic, because the four terms of the balance move in a way that identifies where the constraint bound. Refinery output falls to the bottom of its baseline range while gasoline output, low but well inside its own range, does not, which locates the problem in diesel manufacturing rather than in running the refinery. Demand sits at its baseline, so nothing was rationed and no adjustment came from consumption. Exports collapse and imports reach the top of the baseline range. A system that cannot make as much of a product, and whose consumers want the usual amount of it, has two ways to close the gap: stop selling it abroad and start buying it. Portugal did both, in that order of magnitude.

This is what the trade channel absorbing a shock looks like, and it is the first direct observation in this paper of resilience in the sense of Section 2 being tested. The domestic buffer contributed nothing: output moved the wrong way, which is what a feedstock disruption does. The whole adjustment ran through imports. Whether that counts as a system coping or a system exposed depends on a judgement this paper cannot make, because the terms on which those imports were obtained, and whether they would have been available in a tighter market, are not in the data.

The export figure needs care, because 2022 diesel exports is the single largest disagreement between the two independent trade sources, and the reconciliation identifies the monthly source as the outlier for that cell. Table 9 therefore reports the statistic on both values. The qualitative result is unchanged, since every baseline year exceeds either figure, but the magnitude is not: the standardised deviation is −2.44 on the monthly source and −1.89 on the corroborated national figure. The weaker of the two is the one this paper relies on.

The concentration point matters here in a way it does not elsewhere. A disruption to one feedstock route is a different object in a two-site system than in a one-site system: with two conversion sites, a problem specific to one of them can be partly offset by the other, and with one there is no domestic offset available at any price. Portugal entered 2022 having become a one-site system ten months earlier. The evidence here cannot separate how much of the 2022 movement is the shock and how much is the reduced system meeting it, and that is not a technicality: the two are not independent. A closure changes how a subsequent common shock propagates, so the common-shock account and the closure account are not competing explanations to be weighed against each other but two parts of one causal chain that this design cannot decompose.

9 Prices

9.1 Diagnostics and model choice

Weekly pre-tax prices are the primary measure. The diagnostics are run on log levels, which is the scale the models are estimated on: the short-run specification regresses Δ log PT on Δ log ES, and the error-correction model takes its long-run residual from log PT on log ES. An earlier version of this analysis ran them on the EUR per 1000 litres levels, and the two scales do not agree where it matters: diesel Engle–Granger is 0.088 on the EUR levels and 0.022 on logs, which is the difference between fitting an error-correction model and declaring the pair not cointegrated. The log-scale result is the one reported here.

Augmented Dickey–Fuller tests on the log levels do not reject a unit root for diesel in either country: Portugal $p = 0.145$, Spain $p = 0.214$. For gasoline they reject marginally, Portugal $p = 0.041$ and Spain $p = 0.049$, but that verdict is not robust: substituting BIC for AIC in the lag selection moves the Spanish figure to 0.071 and the rejection disappears. Across the six combinations of deterministic term and lag rule tried, no series rejects at one per cent. A levels regression is admitted only on rejection at one per cent rather than five, because the two errors are not symmetric. Regressing one near-integrated series on another is the spurious regression case and its inference is invalid, whereas an error-correction model whose series turn out to be stationary retains a valid stationary regressor and nests the difference specification. Neither product clears that bar.

Both pairs are then cointegrated in logs: diesel Engle–Granger $p = 0.022$ and gasoline $p = 0.000053$. An error-correction model is therefore indicated for both, and estimated in Section 9.4.

9.2 The levels model, and why it is not used

The levels regression was estimated and is retained in the results archive, flagged as not valid for inference for both products. It is reported here because discarding it silently would hide how much specification choice matters, not because it supports any claim.

On that specification the diesel interaction term ES $\times$ post is -0.101 ($p < 0.001$): Portuguese prices appear to become *less* responsive to Spanish prices after the transition. The licensed error-correction model gives $+0.529$ ($p < 0.001$) on the same interaction, the opposite sign. The levels model also reports a deterministic trend of -0.015 per week ($p = 0.005$). Both features are what a regression between two integrated series estimated without its cointegrating relationship is expected to produce. No conclusion in this paper rests on it, and the sign reversal is the reason the stationarity bar in Section 9.1 is set where it is.

Table 10: Contemporaneous elasticity of Portuguese to Spanish pre-tax prices, Δ log, HAC(8), $n = 1{,}078$ weeks.

Product	Term	Estimate	Std. error	p
diesel	Δ log ES	0.7348	0.0341	< 0.001
	Δ log ES $\times$ post	0.2724	0.0543	< 0.001
	post	-0.0002	0.0009	0.803
gasoline	Δ log ES	0.7997	0.0593	< 0.001
	Δ log ES $\times$ post	0.0353	0.1373	0.797
	post	-0.0000	0.0011	0.969

9.3 Short-run co-movement

This is the difference-only specification. It is nested in the error-correction models below, which are the ones the diagnostics licence, and it is reported because it is what a reader would obtain without the long-run term and because the two give different short-run elasticities. On this specification the weekly elasticity of Portuguese to Spanish pre-tax diesel prices rises from 0.73 to 1.01; the gasoline interaction is small and indistinguishable from no change. Omitting the disequilibrium term is not innocuous: for diesel it moves the post-transition elasticity from 1.24 to 1.01, because part of the adjustment it leaves out is loaded onto the contemporaneous coefficient.

9.4 Error correction

Both pairs are cointegrated in logs, so for both the short-run coefficient is not the whole result. Table 11 reports the two-step Engle–Granger models. Both long-run relations are close to unit elastic: $\log \mathrm{PT} = 0.279 + 0.956 \log \mathrm{ES}$ for diesel and $\log \mathrm{PT} = 0.148 + 0.974 \log \mathrm{ES}$ for gasoline.

The adjustment speed, the share of a gap against that relation closed each week, more than triples for both products after the transition. For diesel it moves from -0.133 to -0.466, shortening the half-life of a deviation from 4.9 to 1.1 weeks. For gasoline it moves from -0.091 to -0.315, a half-life of 7.3 weeks falling to 1.8. Both changes are precisely estimated.

The two products differ in the other channel. The diesel contemporaneous elasticity rises from 0.713 to 1.242, and the increase is precisely estimated ($p < 0.001$). The gasoline contemporaneous elasticity does not move: its interaction is 0.205 with $p = 0.218$, so the post-transition level of 0.994 is not distinguishable from the pre-transition 0.789.

Linkage to Spain therefore tightened for both fuels, and for both it tightened in the long-run adjustment channel, which a difference-only model cannot observe by construction. For diesel the contemporaneous channel tightened as well. Reporting only the short-run coefficient would have supported the conclusion that gasoline was unaffected, and that conclusion would have been an artefact of the specification.

What this amounts to is a loss of domestic price insulation. Before the transition a gap between the Portuguese and Spanish pre-tax diesel price took about five weeks to half-close; afterwards it takes a little over one. A Portuguese price that departs from the Iberian relation is pulled back roughly three and a half times faster than it used to be, and it departs less in the first place, since the contemporaneous elasticity has risen to 1.242. Portuguese pre-tax diesel prices behave less like the output of a domestic market with its own supply conditions and more like a local expression of an Iberian one.

That reading sits naturally alongside the physical result. A country meeting a larger share of its diesel demand from imports is buying at prices formed in the market those imports come

Table 11: Error-correction models, two-step Engle–Granger, HAC(8), $n = 1{,}078$ weeks.

Product	Term	Estimate	Std. error	p
diesel	Disequilibrium (lagged)	−0.1326	0.0257	< 0.001
	× post	−0.3335	0.0784	< 0.001
	post-period level	−0.4662	0.0741	< 0.001
	Δ log ES	0.7129	0.0396	< 0.001
	× post	0.5289	0.1124	< 0.001
	post-period level	1.2418	0.1052	< 0.001
	post	−0.0071	0.0017	< 0.001
gasoline	Disequilibrium (lagged)	−0.0907	0.0354	0.010
	× post	−0.2241	0.0741	0.003
	post-period level	−0.3148	0.0652	< 0.001
	Δ log ES	0.7889	0.0720	< 0.001
	× post	0.2052	0.1667	0.218
	post-period level	0.9941	0.1503	< 0.001
	post	−0.0018	0.0014	0.204

from, so there is less scope for a domestic supply position to hold the local price apart. The two findings are consistent and their timing coincides, which is the strongest statement the design supports. It is not evidence that the physical change caused the price change. The same tightening would follow from a period in which common cost shocks came to dominate both markets, and 2021–2024 was exactly such a period. Distinguishing the two would need refinery-gate or wholesale prices and an import-origin series, neither of which is observed here.

Gasoline is the useful check on that story. Its contemporaneous elasticity does not move, so whatever tightened the diesel relation did not simply tighten everything. But its adjustment speed does move, by a similar factor, and gasoline is the product whose physical position barely changed. That weakens any account tying the price result directly to the physical one, and it is reported here rather than left out because it is the observation least favourable to the paper's own framing.

Two caveats belong with these estimates. The disequilibrium term is a generated regressor, so the second-stage standard errors are conditional on the first-stage cointegrating estimate; HAC covariance mitigates but does not remove this. And the diesel cointegration verdict rests on $p = 0.022$, which clears five per cent but not one, so the diesel model is the more specification-dependent of the two.

9.5 Price spreads

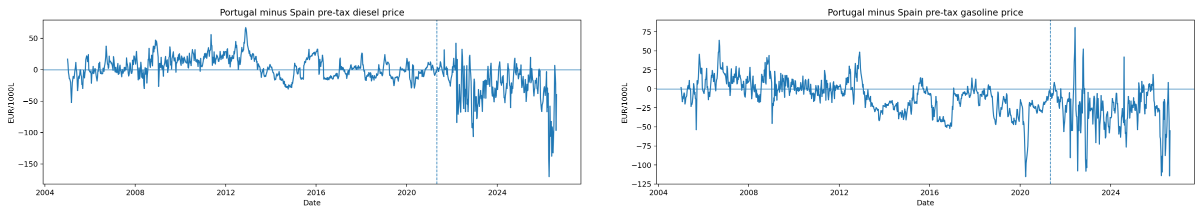

Figure 9: Portugal and Spain weekly pre-tax prices and their spread: diesel (left), gasoline (right).

The mean Portugal-minus-Spain pre-tax diesel spread moves from +4.12 to −26.71 EUR per

1000 litres across the transition, a change of −30.83; the gasoline spread moves by −18.90. These are descriptive.

The two spreads behave differently under test, on 1,079 observations each. The gasoline spread rejects a unit root decisively (ADF $p = 0.0004$) and can be read as fluctuating around a stable level. The diesel spread does not reject ($p = 0.256$), so it must not be read as a disequilibrium measure, and the change in its mean across the transition describes where the series sat rather than a departure from an equilibrium.

This does not contradict Section 9.1, where both pairs are cointegrated. The raw difference PT − ES is the cointegrating combination only if the long-run coefficient is one and the relation holds in levels rather than logs. Neither condition is met: the estimated slopes are 0.956 for diesel and 0.974 for gasoline, and the relation is estimated in logs. A pair can therefore be cointegrated while its unweighted EUR spread wanders, which is what the diesel series does. These spread tests are descriptive companions to the figures above, not a second opinion on the model choice.

10 Portugal–Spain Comparison

10.1 The physical divergence is Portugal-specific

Spain refines the same products and was exposed to the same European energy shock in 2022, so its diesel balance is a useful, if imperfect, reference for what a shock common to the Iberian market did to a neighbouring system. Table 12 sets the two side by side.

The reason to look at Spain is specific. The two countries share a border, a product market and a regulatory environment; they faced the same withdrawal from Russian products on the same timetable; and Spanish prices are already the comparison series in the price models, so the same country carries both halves of the paper. What differs is the event. Portugal closed a refinery in May 2021 and Spain did not. If the post-2021 Portuguese movement were the European shock working through any Iberian refining system, Spain should show something like it. Spain shows nothing like it, which is the whole content of the comparison.

Table 12: Diesel balance ratios, Portugal against Spain.

	Refinery output / demand			Net imports / demand		
Year	PT	ES	PT−ES	PT	ES	PT−ES
2018	1.049	0.899	+0.150	−0.056	−0.046	−0.010
2019	0.905	0.902	+0.003	+0.070	−0.032	+0.102
2020	1.077	0.932	+0.145	−0.084	−0.051	−0.033
2021	0.912	0.847	+0.065	+0.026	+0.009	+0.016
2022	0.867	0.904	−0.037	+0.140	+0.001	+0.139
2023	0.678	0.923	−0.245	+0.262	−0.041	+0.303
2024	0.855	0.898	−0.043	+0.105	−0.040	+0.145

Spanish diesel output covers between 0.85 and 0.93 of Spanish demand in every year shown, with no trend over those years: Spain absorbed 2022 without a change in its physical balance. The comparison is deliberately confined to them. Spain's ratio is not flat across the full panel, falling to 0.667 in 2007 before recovering, so a claim of stability is defensible only over the window shown. Portugal covered more of its own diesel demand than Spain did in 2018, 2020 and 2021, and less from 2022, falling to 0.678 in 2023 while Spain sat at 0.923. Across those years Spanish net imports stay within 0.052 of balance; Portugal reaches 0.262 in 2023.

This does not identify a causal effect, and Spain is not used as a control. A control would have to be a system alike in every relevant respect except the treatment, and two national refining sectors never are. It weakens the common-shock explanation without disposing of it: a divergence appearing in one country and not the other is not by itself sufficient evidence that a shared shock was unimportant, because the two systems differ in ways this paper has not measured. Crude slate, refinery maintenance schedules, inventory positions, demand composition and product-market exposure all differ, and the Russian VGO disruption in particular did not fall on the two countries equally. Establishing that the common shock was second-order would require diagnostics on each of those margins, none of which are estimated here.

Three alternatives remain live, and they are not equally troubling. The first is that Sines was more exposed to the specific feedstock that was withdrawn than the Spanish system was, in which case the divergence is about feedstock sourcing and not about capacity at all. The second is maintenance: an unplanned outage at the one remaining Portuguese site would produce a similar signature and is not observable in annual balances. The third is demand composition, since the two countries recovered from the pandemic on slightly different paths.

The first two are worth separating from the common-shock story proper. Both run through the fact that Portugal had one conversion site by 2022, and having one site is a consequence of the closure. An account in which a common European shock hit Portugal harder because its refining had just been concentrated is not an alternative to the closure explanation; it is a version of it, with the closure acting on the propagation of the shock rather than on the level of output directly. Only the third alternative, demand composition, is genuinely independent of the closure, and the 2022 diesel demand figure sits close to its own baseline, which is weak support for it.

10.2 Balance residuals

The Eurostat panel permits an internal consistency check absent from JODI. Defining the residual as $Q^{\text{refinery}} + M - X - D$, the residuals are large and do not sit on one side of zero. Across 1990–2024 the Spanish gasoline residual ranges from -20.0 to $+15.2$ per cent of demand and the Spanish diesel residual from -14.7 to $+13.9$; the Portuguese ones are smaller but still reach -9.1 and $+12.0$. They reflect stock changes, refinery fuel and statistical differences that the four reported terms do not close. Balance residuals are therefore reported but not interpreted as flows, and no claim in this paper is derived from a residual. Their size is also why the paper infers nothing about stock behaviour from the balance: a residual of this magnitude would swamp any inventory signal it might carry.

11 Robustness

11.1 Event windows

The pre/post comparisons use five pre-event and three post-event years. Repeating them over three window pairs leaves every conclusion unchanged.

Table 13: Pre/post difference around 2021 by window, 2021 excluded.

Product	Outcome	3 pre / 2 post	5 pre / 3 post	8 pre / 3 post
diesel	net imports / demand	+0.224	+0.277	+0.323
diesel	refinery output / demand	−0.238	−0.283	−0.315
gasoline	net imports / demand	+0.333	+0.393	+0.281
gasoline	refinery output / demand	−0.388	−0.424	−0.321

Every estimate keeps its sign across all three windows. The diesel ratios grow monotonically with a longer pre-window, which is expected: a longer baseline reaches further into the pre-hydrocracker regime and widens the measured gap. The gasoline estimates move the other way, shrinking as the pre-window lengthens, because the gasoline surplus was at its widest in the middle of the window rather than immediately before the transition. No conclusion here depends on the window choice.

On the headline five-and-three window, diesel gross import dependence rises from 0.202 to 0.278, a change of 7.6 percentage points; the diesel net-import-to-demand ratio rises from −0.108 to 0.169, a change of 27.7 percentage points; and diesel refinery output coverage falls from 1.083 to 0.800. These are descriptive pre/post differences, not effect estimates.

11.2 Two further robustness questions

Two checks that would otherwise interrupt the argument are reported in Appendix B. The first re-estimates every 2022 result on each of the two trade sources, since the disputed cells fall in that year; one result changes significance and it is marked wherever it appears. The second applies a Benjamini–Hochberg correction across all twenty-four Chow tests, and six survive it.

12 Limitations

1. **2022 confounds the post-closure window.** The Matosinhos transition and the European energy shock fall ten months apart. The monthly model separates them statistically, but COVID-recovery demand, Sines maintenance and the Russian VGO disruption are not separately identified. Two of those run through the concentration the closure produced and are therefore not alternatives to it, as Section 13 sets out; unplanned maintenance is the one that is genuinely independent and the one no series here can rule out.
2. **No causal design.** No control group, instrument or counterfactual is used. Every post-2021 result is an association measured around a documented event. The Spain comparison constrains the common-shock explanation but does not license difference-in-differences: no pre-trend or common-shock diagnostics have been estimated.
3. **DGEG trade covers only 2019–2024.** The 2013 break is corroborated against Eurostat, which is compiled from the DGEG submission rather than independent of it.
4. **Low annual power after 2021.** Three post-event annual observations cannot support a Chow test. The interrupted-trend models carry the annual evidence for 2022, and they impose a common trend that the Chow design does not.
5. **Generated regressor in both error-correction models.** The disequilibrium term is estimated in a first stage, so the reported second-stage standard errors are conditional on it. A single-step or maximum-likelihood estimator would price that uncertainty properly. This now bears on the diesel result as well as the gasoline one, and the diesel model carries the additional caveat that its cointegration verdict clears five per cent but not one.
6. **JODI assessment codes unverified.** All Portuguese observations carry assessment code 1, whose exact semantics have not been confirmed against JODI documentation.
7. **Retail, not wholesale, prices.** The Weekly Oil Bulletin reports consumer prices. The price results describe retail co-movement, not refinery-gate or international wholesale transmission, and are not pass-through in the causal sense.
8. **Balance residuals are not closed.** The four Eurostat terms do not sum to zero and stock changes are not modelled.

13 Conclusions

13.1 What changed in Portugal's dependence profile

Over thirty-five years Portugal's diesel system crossed the same line three times and ended further from self-coverage than it began. It was a net exporter through most of the 1990s, a net importer for the fourteen years from 1999, an exporter again from 2013 after the hydrocracker, and an importer in every year since 2021. By 2023 domestic manufacturing covered 0.678 of diesel demand and the net-import ratio stood at 0.262: the weakest coverage and the heaviest dependence anywhere in the window.

Read against the four senses of resilience set out in Section 2, the profile changed on all four and in the same direction. The domestic buffer thinned: the diesel coverage ratio fell below its 1990s level. Import exposure rose on both measures, in volume and in net position. Manufacturing concentrated from two sites to one. Price integration tightened, on both the contemporaneous and the adjustment channel. Nothing moved the other way.

Gasoline did not follow, and the reason is the one given in Section 5.4 rather than anything about how the two products were treated. Gasoline entered every one of these events with output far above domestic demand and left with output still far above it. The same proportional loss of capacity that pushed diesel across a line left gasoline comfortably on one side of it. That is the clearest practical lesson available here: what determined exposure was not the size of the shock but how much margin the product had before it arrived.

13.2 The best explanation available

Putting the pieces together, the account this paper can defend is one of interacting causes rather than a single one, and the interaction is what makes the period distinctive.

Portugal's diesel position was never a fixed property of the country. It was the running balance between a demand structure that had dieselised faster than the refining system could follow and a refining configuration that was periodically adjusted to catch up. The 2013 hydrocracker was such an adjustment, and it worked: it moved the country back into diesel surplus for most of the following decade. The 2021 closure was an adjustment in the other direction, and it did more than remove capacity. It concentrated what remained in a single site, so that both the volume of national diesel manufacturing and its feedstock exposure came to rest in the same place.

Ten months later a shock arrived that was specific to that feedstock. In the earlier configuration it would have hit one of two plants. In the new one it hit all of Portuguese refining, and the balance shows the consequence exactly: diesel manufacturing at the bottom of its baseline range, gasoline manufacturing inside its own, demand close to baseline, exports at the zeroth percentile and imports at the hundredth. The system met the shock, but it met it entirely through the trade channel, because after May 2021 there was no other channel to meet it through.

The import series is worth reading carefully here, because it does not single 2022 out. Diesel imports were higher in 2021, the closure year, than in 2022, and higher again in 2023 than in either. What 2022 contributes is not the largest import volume in the window but the clearest picture of the mechanism, because it is the year in which the four terms of the balance move against each other in a way that identifies where the constraint bound.

On that reading the closure and the shock are not competing explanations for 2022 and it is a mistake to ask which mattered more. The closure determined how a subsequent shock would propagate; the shock determined when that would be tested. Gasoline is the control the data supplies for this: it went through the same closure at the same plant and shows almost none of it, because it entered with a margin that diesel did not have.

13.3 What can and cannot be inferred

The causal question is open and this design cannot close it. The monthly model finds both phases independently associated with lower refinery output and higher net import dependence, with the 2022 shift the larger of the two, and that is as far as the evidence goes. It is worth being clear that no better estimator would close the gap either, for the reason given above: the decomposition being asked for is not a property the physical system has.

Nor can the paper say how much of the post-2021 level is permanent. Coverage recovers from 0.678 in 2023 to 0.855 in 2024, which is consistent with 2023 having been the trough of an adjustment and equally consistent with a system that now runs closer to its import channel and fluctuates around it.

What follows for policy is correspondingly narrow, and narrower than the subject usually invites. The structural exposure measures moved, and they moved together and in one direction, which is a fact about the system that a policy discussion would need to start from. But this paper has not observed stocks, storage, port capacity, supplier diversity or the terms on which the additional imports were obtained, and those are the quantities that determine whether higher import exposure is a vulnerability or simply a different and possibly cheaper way of meeting demand. A country importing a large share of its diesel through deep and well-connected markets may be better placed than one manufacturing more of its own from a single exposed feedstock route. Nothing here can tell those two states apart, and a reader who takes rising import dependence as *prima facie* evidence of reduced security is going beyond what is shown.

The price finding carries the same limit in a different form. Portuguese pre-tax diesel prices now track Spanish ones more closely and revert to the Iberian relation about three and a half times faster. That is co-movement between two retail series, not pass-through, and the specification controls for nothing the two markets have in common. Nothing here speaks to refinery-gate or wholesale prices, which are not observed at all.

13.4 What evidence would settle it

The obvious constraint is time. Three post-closure annual observations cannot support a Chow test, and several results in this paper are weak precisely because the post-event window is short. A few more years would let the annual designs speak, and would answer the question left open above about whether 2023 was a trough or a level.

Three other kinds of evidence would do more than waiting. A monthly series of Sines throughput and unplanned outages would separate a feedstock disruption from a maintenance event, which annual balances cannot distinguish and which is the most troubling of the remaining alternatives to the account here. An import-origin series would show whether the additional diesel came from the same markets that set the Spanish price, which is the mechanism the price result is consistent with but does not demonstrate. And refinery-gate or wholesale quotations would let the price question be asked as pass-through rather than as co-movement, which is the form in which it is worth asking.

Failing all of those, the cleanest natural experiment would be a comparable single-refinery closure elsewhere in Europe at a date not adjacent to a continent-wide supply shock. The identification problem in this paper is not a shortage of data about Portugal. It is that the two events fall ten months apart, and no amount of Portuguese data will move them.

Claim–Evidence Matrix

Claim	Where shown	Level	Remaining caveat
Sines hydrocracker entered commercial production in 2013	Section 1	documented event	Corporate disclosure, not outcome data
Matosinhos refining ceased May 2021	Section 1	documented event	Announcement date differs from closure date
2022 Russian product and VGO disruption	Section 1	documented event	Establishes timing, not magnitude
Diesel demand averaged 4,522 kt, output 4,348 kt	Section 5.1	descriptive	Window means, not a model
Portugal was a net diesel exporter in every year 2013–2018, and in 2020	Tables 2, 3	descriptive	Ratio is not self-sufficiency; 2019 is a net-import year
Gasoline is a net export product in every year but 1993	Tables 2, 3	descriptive	Same caveat; 1993 net-import ratio +0.056
Balance at each event year	Table 4	descriptive	2021 marked transition
Refining regime by year	Table 4	documented	Regime label, not capacity
Diesel net-import ratio breaks at 2013	Table 5	structural break	Diagnostic, not a causal estimator
No Chow break detected at 2022	Table 5	structural break	Three post-event years; low power
2000 break is exploratory, not pre-specified	Table 5	association	Added after extending the panel; no conclusion rests on it
Interrupted trends do detect 2022	Table 6	association	Imposes a common trend; 2022 export rows depend on disputed cells
Diesel 2022 annual results do not depend on the disputed cell	Table 15	association	Annual panel uses the corroborated source throughout
No 2022 gasoline annual result reaches significance	Table 15	—	Weak on either source; not relied on
Monthly phase means shift after May 2021	Table 7	descriptive	No seasonality control
Refinery output sits −102.89 kt/month below the pre-closure counterfactual in the transition	Table 8	association	Quote with its phase-trend term
In the stress phase it sits −148.25 kt/month below the same counterfactual	Table 8	association	Not incremental to the transition shift; confounded with demand recovery
Monthly models cover 276 months, 2002–2024	Section 3.2	descriptive	JODI begins in 2002, later than the annual window
2022 diesel exports at the 0th baseline percentile	Table 9	descriptive	Sources disagree on this cell; both reported
Diesel log prices are non-stationary; gasoline rejects only marginally	Section 9.1	descriptive	ADF only; the gasoline verdict changes with the lag rule
Both pairs are cointegrated in logs, diesel $p = 0.022$ and gasoline $p = 0.000053$	Section 9.1	descriptive	Engle–Granger, single specification; diesel clears five per cent but not one
EUR-levels model gives the opposite sign and is not used	Section 9.2	—	Retained as a specification warning only

Claim	Where shown	Level	Remaining caveat
Diesel contemporaneous elasticity rises 0.7129 to 1.2418	Tables 11, 10	association	Retail, not wholesale, prices; 1.01 in the difference-only model
Gasoline contemporaneous elasticity unchanged	Table 11	association	Interaction $p = 0.218$; not the whole picture, see the adjustment speed
Both adjustment speeds more than triple	Table 11	association	Generated-regressor standard errors
PT–ES diesel spread moves −30.8 EUR/1000L	Section 9.5	descriptive	That spread is non-stationary; not a disequilibrium measure
Diesel spread is not mean-reverting; the gasoline spread is	Section 9.5	descriptive	Diesel ADF $p = 0.256$, gasoline $p = 0.0004$
Spanish diesel balance is flat while Portugal's falls	Table 12	descriptive	Weakens but does not dispose of the common-shock account
Eurostat balance residuals are not closed	Section 10.2	descriptive	Stock changes not modelled
Pre/post differences hold across three windows	Table 13	descriptive	Magnitudes vary with baseline length
Headline pre/post dependence change	Section 11.1	descriptive	Not an effect estimate
JODI coverage is complete for every year	Table 1	descriptive	Assessment-code semantics unverified
Sources agree on 89% of full-window trade cells	Section B.1	descriptive	Eurostat is DGEG-derived
Sources and retrieval points are frozen	Section 3.1	documented	Snapshot of one vintage
Hypotheses were pre-specified	Section 1	documented	Recorded before estimation
No causal effect of the closure is claimed	—	—	2022 is not separable by this design

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A Data and Code Availability

All data underlying this paper are public and are listed in the References. The analysis code, the derived tables behind every figure and table above, and a document recording the provenance of each one are available in the project repository, reference [9].

Derived results are archived with a checksum manifest that accounts for every file in the archive, so the tables and figures reproduced here can be verified against the exact inputs used to produce them. Eight automated readiness checks are applied before any figure is reported: they verify archive completeness, annual coverage of the monthly source, validity of the monthly panel and its models, availability of the cross-country balance, completeness of the price diagnostics including the recorded model-family verdict, and source reconciliation. The reconciliation check additionally requires every out-of-tolerance cell to be named with a reason, so a later data vintage cannot widen the gap between sources silently. All eight pass for this paper.

B Source Reconciliation, Sensitivity and Multiplicity

This appendix holds the material that establishes the evidence is sound: how the sources are compared, what happens to the results when a disputed cell is replaced, and how the multiple testing is corrected. It is placed here rather than in the body because it answers a question about the evidence rather than about Portugal, and every claim it supports is stated in the body with a pointer back to this appendix.

B.1 Source reconciliation

JODI and DGEG are the only genuinely independent pair. Across the 24 overlapping 2019–2024 trade cells, 66.7% agree within tolerance. Over the full window, JODI and Eurostat agree on 89% of 80 cells, with only three of the nine divergences occurring before 2019.

A cell is flagged only when it breaches both the absolute and the percentage tolerance. Applying them as alternatives makes a 25 kt floor bind on series of roughly 1,600 kt, an effective tolerance of 1.6%, which would flag agreement as close as 2.1% as failure. Each of the eight remaining divergent cells is enumerated in the analysis configuration, together with the series Eurostat identifies as the outlier: six where JODI is out of line, two where the DGEG workbook is. An unlisted divergence fails the readiness gate however few cells it touches.

B.2 Sensitivity to the disputed trade cells

Two 2022 trade cells are disputed between the two independent sources: diesel exports, where the monthly source reports 331.0 kt against 622.2 kt nationally, and gasoline exports, where it reports 1,346.0 kt against 1,501.7 kt.

Moving the annual panel to Eurostat changes what that disagreement can affect. The annual results are now computed from the national submission, which already carries the corroborated value for the diesel cell, so the disputed monthly figure does not enter the annual arm at all. Table 15 records this: the diesel rows are identical under both sources because there is nothing to swap. The disagreement now bears only on the monthly arm and on the JODI cross-check, not on the annual conclusions.

The diesel results are unaffected by the swap, for the reason just given, and both remain significant at conventional levels.

Table 15: 2022 interrupted-trend level changes, computed on each trade source.

		Primary source		Corroborated	
Product	Outcome	Estimate	p	Estimate	p
diesel	exports	−781.7	0.022	−781.7	0.022
diesel	net imp./dem.	+0.201	0.031	+0.201	0.031
gasoline	exports	−270.9	0.121	−202.8	0.149
gasoline	net imp./dem.	+0.307	0.095	+0.253	0.111

The gasoline results are weak on either source. Neither the export level change nor the net-import-ratio shift reaches five per cent under the primary values, and swapping to the corroborated ones reduces both further. No 2022 gasoline result is relied on anywhere in this paper. The gasoline evidence for that year rests on the monthly design of Section 6 rather than on the annual models.

B.3 Multiplicity

Twenty-four Chow tests are run and Benjamini–Hochberg adjustment is applied across all of them, not to a selected subset. Six survive at conventional levels. The monthly models are not included in that adjustment and their p-values should be read as unadjusted; the phase terms that carry the conclusions, on refinery output and on the net-import ratio, have $p \leq 0.007$, inside any reasonable correction for the twelve terms reported. The import-level terms are weaker and are not relied on: the Matosinhos import shift is significant only at the ten per cent level.